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Thermodynamic assessments of the Ag–Y and Sc–Y systems

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ABSTRACT

The phase diagrams and thermodynamic properties in the Ag–Y and Sc–Y binary systems have been assessed by using the CALPHAD (*Cal*culation of *Phase D*iagrams) method on the basis of the experimental data including the thermodynamic properties and phase equilibria. The Gibbs free energies of the liquid, bcc, fcc, and hcp phases were described by the subregular solution model with the Redlich–Kister equation, and those of the intermetallic compounds ($Ag_{51}Y_{14}$, Ag_2Y and AgY phases) in these two binary systems were described by the sublattice model. The thermodynamic parameters of the Ag–Y and Sc–Y binary systems were obtained, and an agreement between the calculated results and experimental data was obtained in each binary system.

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1. Introduction

Sc-based alloys have been used as aero-structures materials [1], and Sc addition on wrought alloys has positive effects on weldability and welding properties [2]. The phase equilibria in the Sc-based alloy systems are of importance for design of new Sc-based alloys. The CALPHAD method, which is a powerful tool to decrease cost and time during development of materials [3], effectively provides a clear guideline for material design. As a result, it is of great essence to establish the thermodynamic database for the Sc-based alloy systems. In the present paper, as a part of thermodynamic assessment in the Sc–Ag–Y system, the thermodynamic descriptions for the phase diagrams in the Ag–Y and Sc–Y binary systems were carried out by combining thermodynamic models with the experimental data in the framework of the CALPHAD method.

2. Experimental information

2.1. The Ag–Y system

The Ag–Y system is consisted of three solid solution phases (fcc (Ag), αY and βY phases) and three intermetallic compounds ($Ag_{51}Y_{14}$, Ag_2Y and AgY phases). Based on microscopy, X-ray diffraction, thermal analysis and hardness measurements, the phase diagram in the Ag–Y system was investigated by Gebhardt et al. [4], who reported three stoichiometry compounds of Ag_xY (melting point, 940 °C), Ag_2Y(melting point, 960 °C) and AgY (melting point, 1160 °C). Gebhardt et al. [4] also determined the maximum solid solubility of Y in the fcc (Ag) solution phase to be less than 1 at.% at 775 °C and four eutectic reactions: L ↔ (Ag) + $Ag_{51}Y_{14}$ at 11.5 at.% Y and 775 °C, L ↔ $Ag_{51}Y_{14}$ + Ag_2Y at 29 at.% Y and 900 °C, L ↔ Ag_2Y + AgY at 35 at.% Y and 945 °C, and L ↔ AgY + αY at 72.5 at.% Y and 885 °C. The allotropic transformation of αY ↔ βY at 1478 °C was not indicated in Ref. [4].

Later, Gschneidner et al. [5] studied the phase diagram in the Ag-rich region, and determined the Ag-rich compound to be the $Ag_{51}Y_{14}$ phase. Mcmasters et al. [6] reported that the $Ag_{51}Y_{14}$ compound has the hexagonal structure ($Ag_{51}Gd_{14}$-type) by X-ray powder diffraction methods. Gschneidner et al. [5] also determined the temperature of the eutectic reaction (L ↔ (Ag) + $Ag_{51}Y_{14}$) to be 799 ± 3 °C and the terminal solid solubility in fcc (Ag) solution phase to be 1.31 at.% Y.

The major difference between the results of Gebhardt et al. [4] and Gschneidner et al. [5] is the maximum solid solubility of Y in the fcc (Ag) solution phase and the eutectic temperature. The solid solubility and the eutectic temperature reported by Gschneidner et al. [5] were accepted in the Ag–Y binary phase diagram compiled by Gschneidner and Calderwood [7] with some modifications from the experimental data reported by Gebhardt et al. [4]. The phase diagram of the Ag–Y system reviewed by Gschneidner and Calderwood [7] is shown in Fig. 1.

In addition, Fitzner et al. [8] measured the standard enthalpies of formation of the $Ag_{51}Y_{14}$ and AgY phases by direct high-temperature reaction calorimetry, and also determined the enthalpies of mixing of liquid phase at 1200 °C. Meschel and Kleppa [9] measured the standard enthalpy of formation of the Ag_2Y phase by direct high-temperature reaction calorimetry. Ivanov and Lukashenko [10–11] measured the activities of components and Gibbs free energies of formation of liquid phase at 1346 and 1363 K by the Knudsen effusion method.

2.2. The Sc–Y system

The Sc–Y system is consisted of two reciprocal solution phases (αSc and αY, βSc and βY phases). Based on thermal analysis, optical pyrometer and X-ray diffraction methods, the phase diagram in the Sc–Y system was investigated by Beaudry and Daane [12], who indicated the minimum solidus at 50 at.% Y and 1365 °C, and the minimum transition temperature from the hcp phase to the bcc phase at 43 at.% Y and 1175 °C. But, the melting temperatures and the β ↔ α transformation temperatures did not agree with the values of the pure metals in Ref. [13]. Consequently, the Sc–Y binary phase diagram was compiled by Gschneidner and Calderwood [14] with

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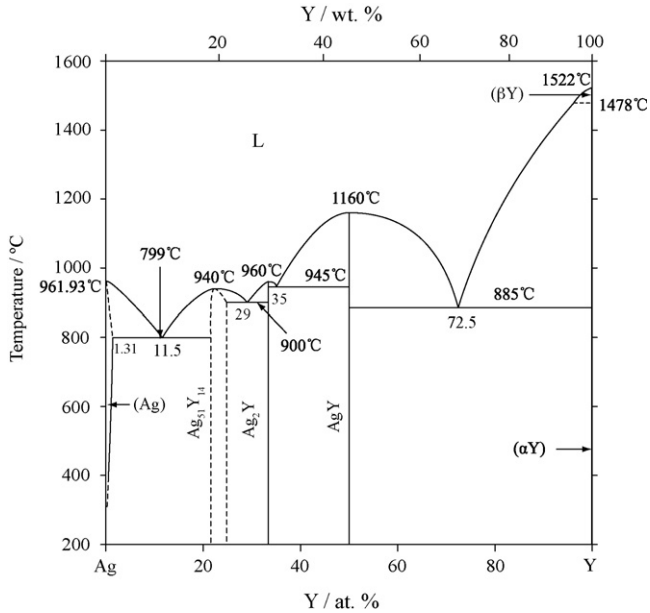

Fig. 1. The phase diagram of the Ag–Y system reviewed by Gschneidner and Calderwood [7].

some modifications from the experimental data reported by Beaudry and Daane [12]. The phase diagram of the Sc–Y system reviewed by Gschneidner and Calderwood [14] is shown in Fig. 2.

In addition, Niessen et al. [15] predicted that the enthalpies of solution at infinite dilution: Sc (liquid) in Y (liquid) is 3 kJ/mole and Y (liquid) is 4 kJ/mole. However, no temperature dependence was given in the work of Niessen et al. [15].

3. Thermodynamic models

The information of stable solid phases and the used models in the Ag–Y and Sc–Y binary systems are listed in Table 1.

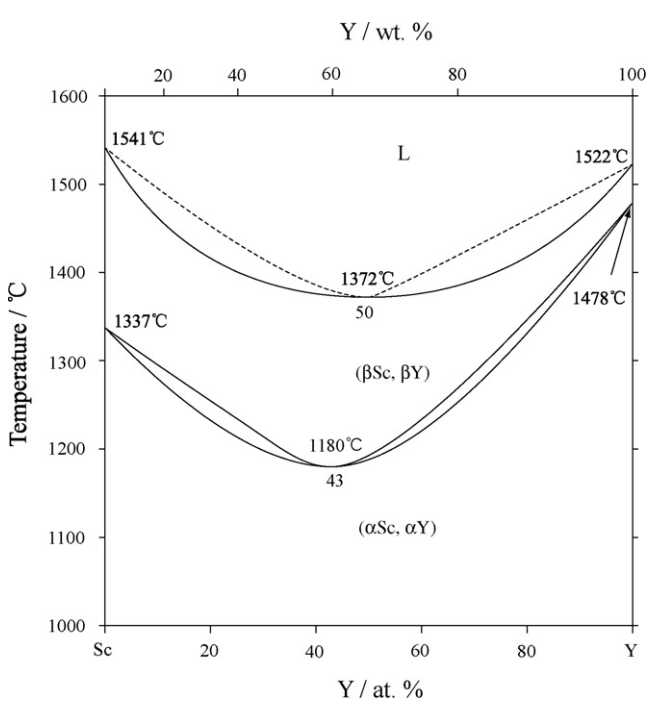

Fig. 2. The phase diagram of the Sc–Y system reviewed by Gschneidner and Calderwood [14].

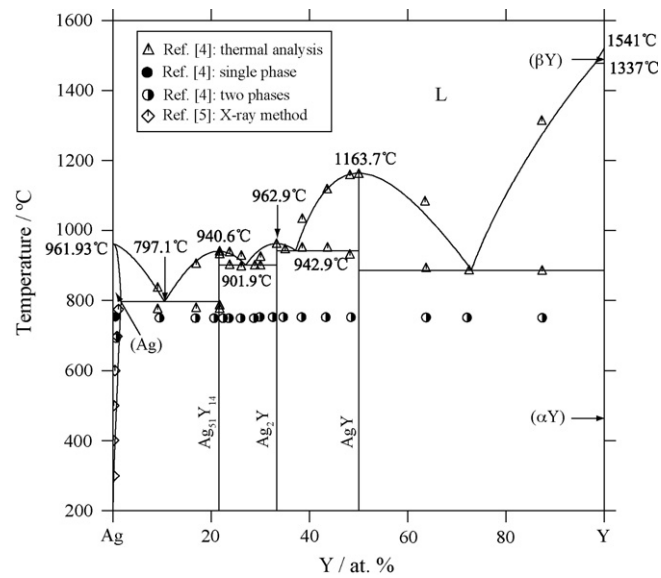

Fig. 3. The calculated phase diagram of the Ag–Y system compared with the experimental data [4–5].

3.1. Solution phases

In the Ag–Y and Sc–Y binary systems, Gibbs free energies of the liquid, fcc, hcp and bcc phases are described by the subregular solution model, as follows:

$$G_m^{\phi} = \sum_{i=\mathrm{M,Y}} {}^0G_i^{\phi} x_i + RT \sum_{i=\mathrm{M,Y}} x_i \ln x_i + \Delta^E G_m^{\phi}, \quad (1)$$

where ${}^0G_i^{\phi}$ is the Gibbs free energy of the pure component i in the respective reference state with the ϕ phase, which is taken from the SGTE pure element database [16]; M means the element Ag or Sc; x_i denotes the mole fraction of the component i; R is the gas constant; T is the absolute temperature; and the term $\Delta^E G_m^{\phi}$ presents the excess Gibbs free energy, which is expressed in the Redlich–Kister polynomial [17] as

$$\Delta^E G_m^{\phi} = x_{\mathrm{M}} x_{\mathrm{Y}} \sum_{m=0}^{n} {}^mL^{\phi}(x_{\mathrm{M}} - x_{\mathrm{Y}})^m, \quad (2)$$

$${}^iL_{\mathrm{M,Y}}^{\phi} = a + bT, \quad (3)$$

where ${}^iL_{\mathrm{M,Y}}^{\phi}$ is the binary interaction parameter, and the coefficients of a and b are constants to be evaluated on the basis of available experimental data.

Table 1
The stable solid phases and the used models in the Ag–Y [7] and Sc–Y [14] systems.

System	Phase	Strukturbericht designation	Prototype	Modeling phase	Used models
Ag–Y	(Ag)	A1	Cu	fcc	SSM
	$Ag_{51}Y_{14}$	–	$Ag_{51}Gd_{14}$	$(Ag)_{51}(Y)_{14}$	SM
	Ag_2Y	C11	$MoSi_2$	$(Ag)_2(Y)$	SM
	AgY	B2	CsCl	(Ag)(Y)	SM
	(αY)	A3	Mg	hcp	SSM
	(βY)	A2	W	bcc	SSM
Sc–Y	(αSc)	A3	Mg	hcp	SSM
	(βSc)	A2	W	bcc	SSM
	(αY)	A3	Mg	hcp	SSM
	(βY)	A2	W	bcc	SSM

Note: SSM: subregular solution model; SM: sublattice model.

Table 2
Thermodynamic parameters in the Ag–Y system assessed in the present work.

System	Parameters in each phase (J/mol)
Ag–Y	Liquid phase, format (Ag, Y) ${}^{0}L^{Liq}_{Ag,Y} = -111,625 + 23.155T$ ${}^{1}L^{Liq}_{Ag,Y} = -23,197 - 18.262T$ ${}^{2}L^{Liq}_{Ag,Y} = 35,266 - 31.164T$ ${}^{3}L^{Liq}_{Ag,Y} = 69117 - 38.98T$ fcc phase, format (Ag, Y) ${}^{0}L^{fcc}_{Ag,Y} = 391,243 - 3.8T$ ${}^{1}L^{fcc}_{Ag,Y} = -450,000 - 39T$ bcc phase, format (Ag, Y) ${}^{0}L^{bcc}_{Ag,Y} = 30,000$ hcp phase, format (Ag, Y) ${}^{0}L^{hcp}_{Ag,Y} = 40,000$ AgY phase, format $(Ag)_{0.5}(Y)_{0.5}$ ${}^{0}G^{AgY}_{Ag:Y} = 0.5 \times {}^{0}G^{fcc}_{Ag} + 0.5 \times {}^{0}G^{hcp}_{Y} - 29,995 + 1.848T$ Ag_2Y phase, format $(Ag)_{0.667}(Y)_{0.333}$ ${}^{0}G^{Ag_2Y}_{Ag:Y} = 0.667 \times {}^{0}G^{fcc}_{Ag} + 0.333 \times {}^{0}G^{hcp}_{Y} - 27,513 + 0.557T$ $Ag_{51}Y_{14}$ phase, format $(Ag)_{0.785}(Y)_{0.215}$ ${}^{0}G^{Ag_{51}Y_{14}}_{Ag:Y} = 0.785 \times {}^{0}G^{fcc}_{Ag} + 0.215 \times {}^{0}G^{hcp}_{Y} - 22,981 + 0.774T$

3.2. Stoichiometric intermetallic compounds

Three compounds of the $Ag_{51}Y_{14}$, Ag_2Y and AgY phases in the Ag–Y binary system are treated as stoichiometric phases, because no reliable experimental data about the homogeneity range of these compounds are available. The Gibbs free energy per molar of formula unit Ag_pY_q can be expressed, as follows:

$$G^{Ag_pY_q}_{Ag:Y} = p^0G^{SER}_{Ag} + q^0G^{SER}_{Y} + \Delta^0G^{Ag_pY_q}_{f}, \quad (4)$$

where the $\Delta^0G^{Ag_pY_q}_{f}$ indicates the standard Gibbs free energy of formation of the stoichiometric compound from pure elements, which is described as

$$\Delta^0G^{Ag_pY_q}_{f} = a' + b'T \quad (5)$$

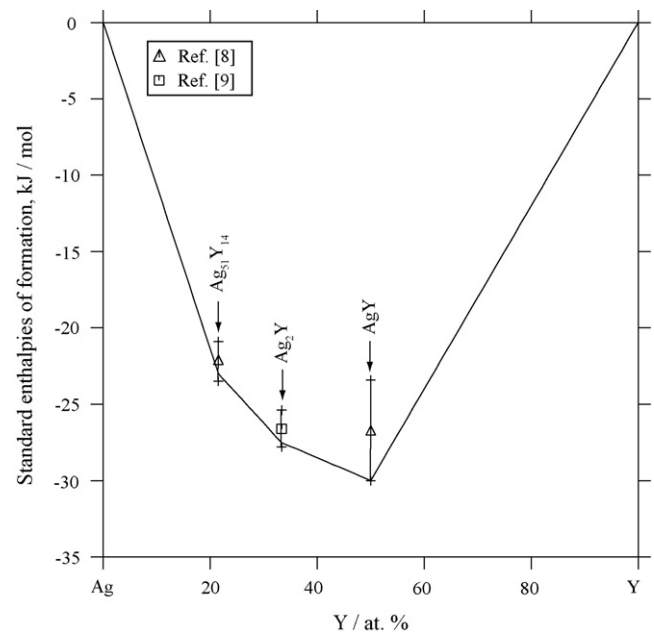

Fig. 4. The calculated standard enthalpies of formation with reference states of Ag (fcc) and αY (hcp) in the Ag–Y system compared with the experimental data [8–9].

where the parameters of a' and b' are evaluated in the present work.

4. Optimized results and discussion

The optimization of the thermodynamic parameters was carried out by using the PARROT program in the Thermo-Calc software [18], which can handle various kinds of experimental data. The experimental data of the phase diagram and thermodynamic properties were used as input to the program. Each piece of selected informa-

Table 3
Special points of the Ag-Y system.

Invariant reaction	Reaction type	Composition at.% Y			Temperature (°C)	Reference
L ↔ βY	Melting		100		1522	[7]
			100		1509	[4]
			100		1522	This work
βY ↔ αY	Polymorphic		100		1478	[7]
			100		1478	This work
L ↔ αY + AgY	Eutectic	72.5	100	50.0	885	[4]
		72.9	100	50.0	887.1	This work
L ↔ AgY	Congruent		50.0		1160	[4]
			50.0		1163.7	This work
L ↔ AgY + Ag_2Y	Eutectic	~35	50.0	33.3	945	[4]
		37.2	50.0	33.3	942.9	This work
L ↔ Ag_2Y	Congruent		33.3		960	[4]
			33.3		962.9	This work
L ↔ Ag_2Y + $Ag_{51}Y_{14}$	Eutectic	~29	33.3	21.5	~900	[4]
		27	33.3	21.5	901.9	This work
L ↔ $Ag_{51}Y_{14}$	Congruent		~22.6		940	[4]
			21.5			[5]
			21.5		940.6	This work
L ↔ $Ag_{51}Y_{14}$ + (Ag)	Eutectic	11.5	~22.6	<1.0	775	[4]
		11.5	21.5	1.31	799 ± 3	[5]
		10.5	20.0	1.6	797.1	This work
L ↔ (Ag)	Melting		0		960	[4]
			0		961.9	[7]
			0		961.9	This work

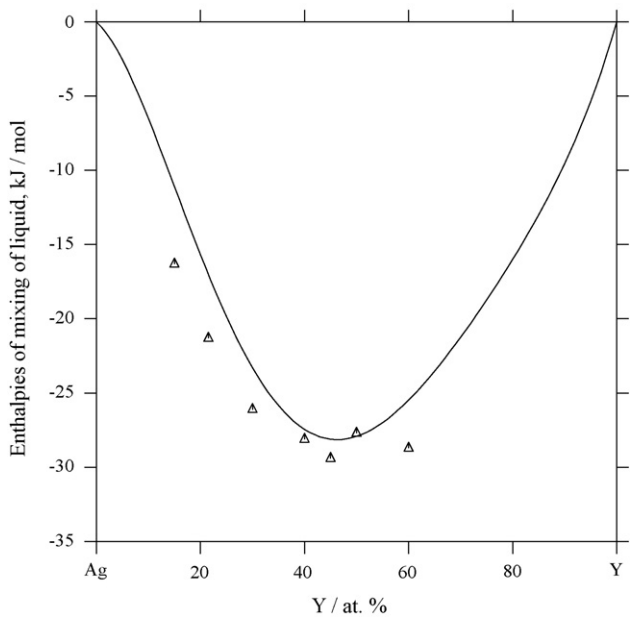

Fig. 5. The calculated enthalpies of mixing of liquid phase at 1200 °C with reference states of pure elements of Ag (liquid) and Y (liquid) in the Ag–Y system compared with the experimental data [8].

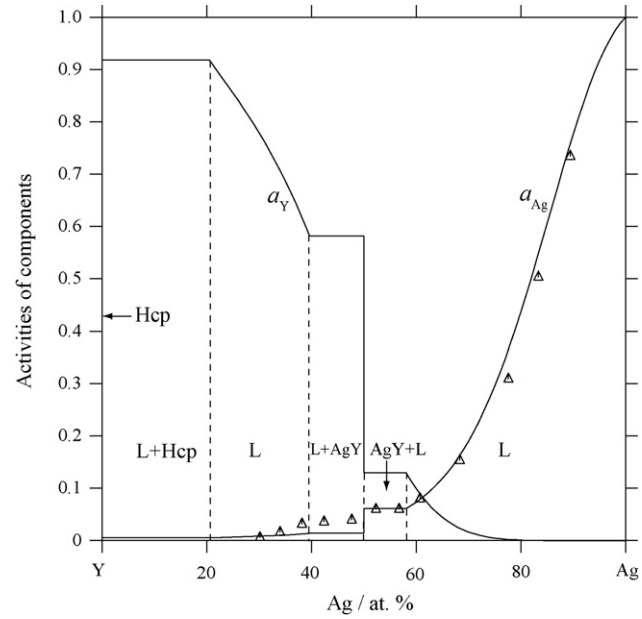

Fig. 7. The calculated activities of components at 1363 K in liquid Ag–Y system with reference states of pure elements of Ag (liquid) and Y (liquid) compared with the experimental data [11].

tion was given a certain weight based on the importance of data, and changed by trial and error during the assessment, until most of the selected experimental information was reproduced within the expected uncertainty limits.

4.1. The Ag–Y system

The calculated Ag–Y phase diagram compared with the experimental data is shown in Fig. 3. It is seen that the calculated results are in reasonable agreement with the experimental data [4–5]. The calculated solid solubility of Y in the fcc (Ag) solution phase is 1.6 at.% Y, which is close to the experimental data (1.31 at.% Y) [5], and this calculated results is reasonable. A set of complete self-consistent thermodynamic parameters describing the Gibbs free energy of each phase in the Ag–Y system are given in Table 2, and the calculated compositions and temperatures for the invariant reactions compared with the selected experimental data are listed in Table 3.

The calculated standard enthalpies of formation with reference states of the Ag (fcc) and αY (hcp) phases compared with the experimental data are presented in Fig. 4, and a comparison of enthalpies of mixing of liquid phase at 1200 °C between the calculated and experimental results is shown in Fig. 5, where there are small discrepancies between the calculated and experimen-

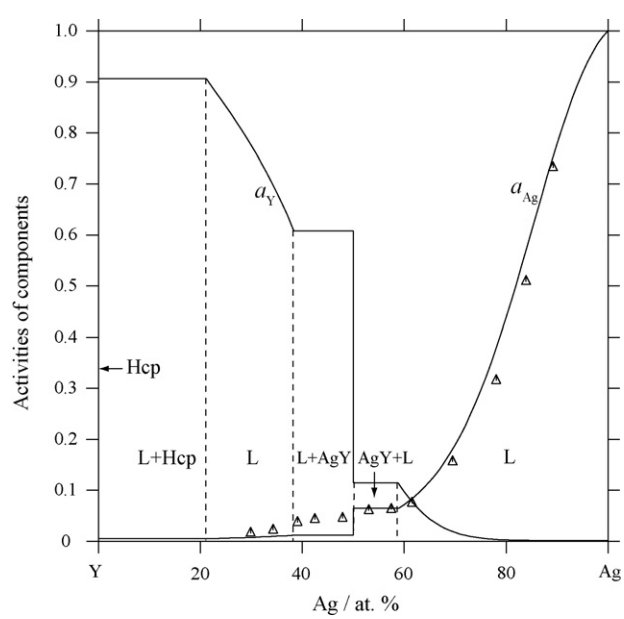

Fig. 6. The calculated activities of components at 1346 K in liquid Ag–Y system with reference states of pure elements of Ag (liquid) and Y (liquid) compared with the experimental data [10].

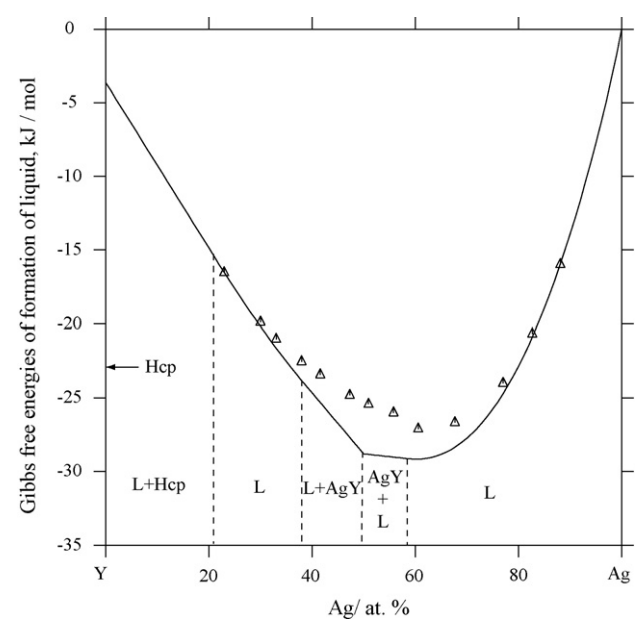

Fig. 8. The calculated Gibbs free energies of formation of liquid at 1346 K with reference states of pure elements of Ag (liquid) and Y (liquid) compared with the experimental data [10].

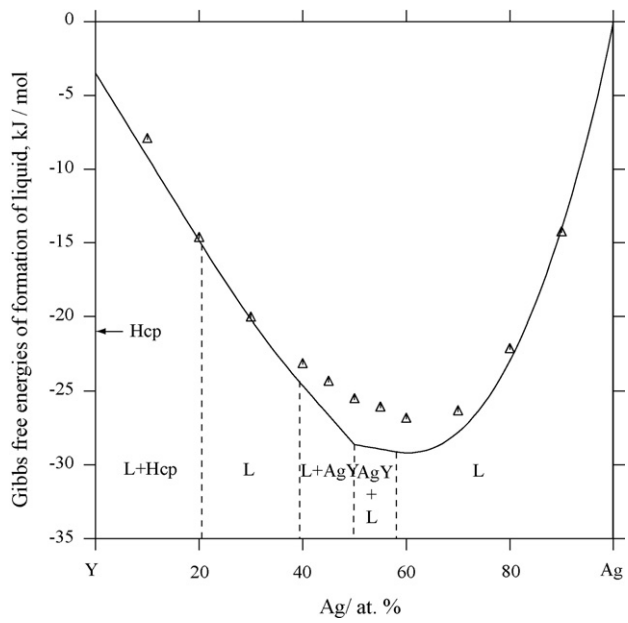

Fig. 9. The calculated Gibbs free energies of formation of liquid at 1363 K with reference states of pure elements of Ag (liquid) and Y (liquid) compared with the experimental data [11].

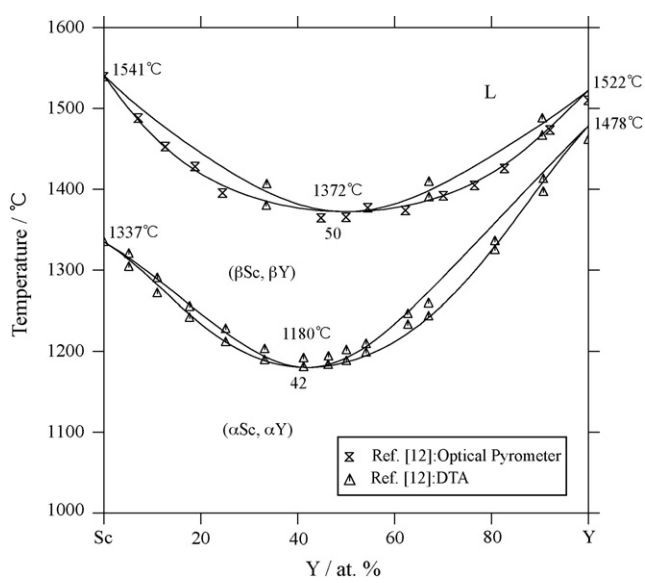

Fig. 10. The calculated phase diagram of the Sc–Y system compared with the experimental data [12].

Table 4
Thermodynamic parameters in the Sc–Y system assessed in the present work.

System	Parameters in each phase (J/mol)
Sc–Y	Liquid phase, format (Sc, Y) ${}^{0}L^{\mathrm{Liq}}_{\mathrm{Sc,Y}} = 13,702 - 1.86T$ ${}^{1}L^{\mathrm{Liq}}_{\mathrm{Sc,Y}} = -0.5T$ ${}^{2}L^{\mathrm{Liq}}_{\mathrm{Sc,Y}} = 5000 - 0.341T$ bcc phase, format (Sc, Y) ${}^{0}L^{\mathrm{bcc}}_{\mathrm{Sc,Y}} = -1032 + 9.784T$ ${}^{2}L^{\mathrm{bcc}}_{\mathrm{Sc,Y}} = 1598 + 0.508T$ hcp phase, format (Sc, Y) ${}^{0}L^{\mathrm{hcp}}_{\mathrm{Sc,Y}} = 1800 + 9.523T$ ${}^{2}L^{\mathrm{hcp}}_{\mathrm{Sc,Y}} = -488 + 1.112T$

Table 5
Special points of the Sc–Y system.

Invariant reaction	Reaction type	Composition at.% Y	Temperature (°C)	Reference
L ↔ βSc	Melting	0	1539	[12]
		0	1541	[13,14]
		0	1541	This work
βSc ↔ αSc	Polymorphic	0	1334	[12]
		0	1337	[13,14]
		0	1337	This work
L ↔ (βSc, βY)	Congruent	50	1365	[12]
		50	1372	[14]
		50	1372	This work
L ↔ βY	Melting	100	1510	[12]
		100	1522	[13,14]
		100	1522	This work
βY ↔ αY	Polymorphic	100	1460	[12]
		100	1478	[13,14]
		100	1478	This work
(βSc, βY) ↔ (αSc, αY)	Congruent	43	1175	[12]
		43	1180	[14]
		42	1180	This work

tal results, which can be accepted by considering that the most of calculated thermodynamic properties including activities, Gibbs free energies, standard enthalpies of formation are in agreement with the experimental data. The calculated activities with experimental data in the liquid binary alloys at 1346 and 1363 K with reference states of pure elements of Ag (liquid) and Y (liquid) are illustrated in Figs. 6 and 7. Figs. 8 and 9 show the calculated Gibbs free energies of formation of liquid phase with experimental data at 1346 and 1363 K with reference states of pure elements of Ag (liquid) and Y (liquid). It is seen that an agreement on the thermodynamic properties is obtained between the calculated results and the experimental data.

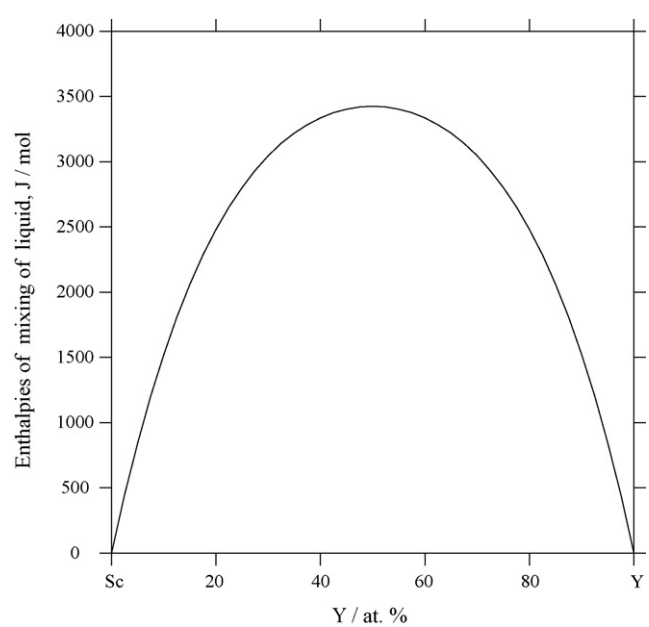

Fig. 11. The calculated enthalpies of mixing of liquid phase at 1600 °C with reference states of pure elements of Sc (liquid) and Y (liquid).

4.2. The Sc–Y system

The calculated Sc–Y phase diagram compared with the experimental data is shown in Fig. 10. It is seen that the calculated phase diagram are in agreement with the experimental data [12,14]. A set of complete self-consistent thermodynamic parameters describing the Gibbs free energy of each phase in the Sc–Y system are given in Table 4, and the calculated compositions and temperatures for the invariant reactions compared with the experimental data are listed in Table 5. The calculated enthalpies of mixing of liquid phase at 1600 °C with reference states of pure elements of Sc (liquid) and Y (liquid) with different components are shown in Fig. 11. It is seen that the enthalpies of mixing of liquid phase are reasonable in agreement with the value predicated by Niessen et al. [15].

5. Conclusions

The phase diagram in the Ag–Y and Sc–Y binary systems were critically assessed by considering the experimental data including phase equilibria and thermodynamic data. A consistent set of thermodynamic parameters has been optimized and most of the experimental information can be satisfactorily reproduced on the basis of the optimized thermodynamic parameters.

Acknowledgements

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